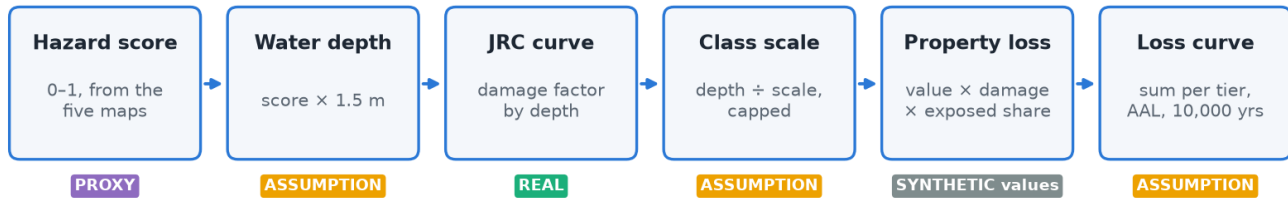


# From flood hazard to loss

How the Xpat Nairobi model turns a hazard score into a property loss and a return-period curve: the published curve it uses, every assumption on top of it, and the arithmetic worked on real rows.



Labels: REAL = published or observed · PROXY = derived, not measured · ASSUMPTION = our modelling choice · SYNTHETIC = invented test data.

| Item                     | Value                                                                  |
|--------------------------|------------------------------------------------------------------------|
| Portfolio                | 600 properties, KES 63.64 bn insured value (SYNTHETIC)                 |
| 1-in-100 loss (scenario) | KES 1.70 bn (2.67% of value)                                           |
| 1-in-250 loss (scenario) | KES 2.65 bn (4.16% of value)                                           |
| Average annual loss      | KES 125.9 m                                                            |
| Configuration            | nairobi-prototype-v0.6, fingerprint acedb441ec75; generated 2026-10-08 |

## 1. The base curve (sourced)

**Source:** Huizinga, J., de Moel, H. & Szewczyk, W. (2017). Global flood depth-damage functions: Methodology and the database with guidelines. European Commission Joint Research Centre, JRC105688. **Table 3-1, average continental damage function for Africa, residential buildings (p.12).**

| Depth (m)     | 0    | 0.5  | 1    | 1.5  | 2    | 3    | 4    | 5    | 6    |
|---------------|------|------|------|------|------|------|------|------|------|
| Damage factor | 0.00 | 0.22 | 0.38 | 0.53 | 0.64 | 0.82 | 0.90 | 0.96 | 1.00 |

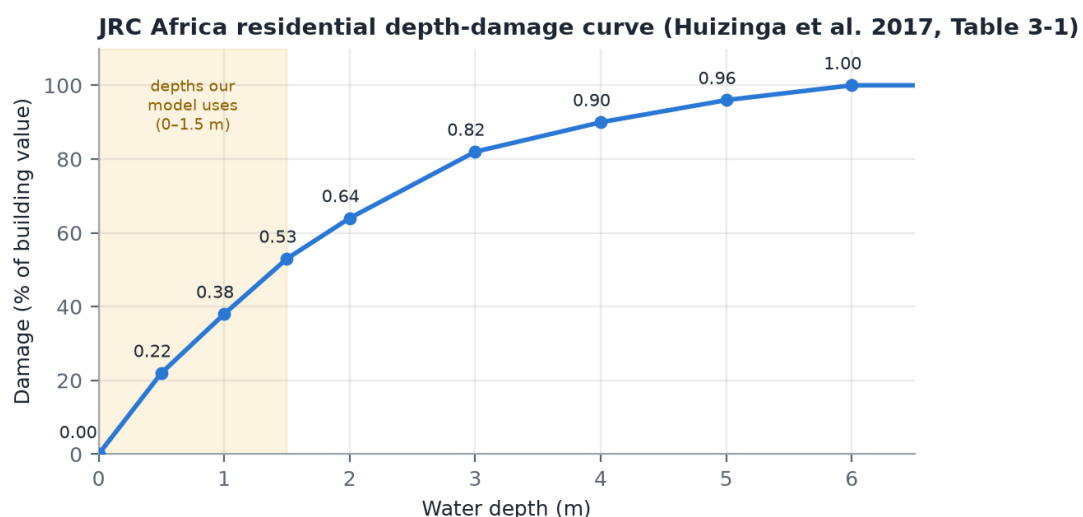


Figure 1. The JRC curve with its published points. Straight lines between points; held at 1.0 beyond 6 m. The shaded band is the depth range our base case can reach.

A damage factor is the share of a building's value destroyed at that depth: 0.53 means 53%. The African curve is built from South African and Mozambican house functions only; no Kenya-specific curve exists publicly, which the brief acknowledges. The curve points are labelled **REAL**.

## 2. Step 1 — hazard score to water depth (ASSUMPTION)

The hazard maps give a 0–1 relative susceptibility score, not a depth. The curve needs metres, so the model uses **depth = score × 1.5 m**. A score of 1 counts as 1.5 m of water.

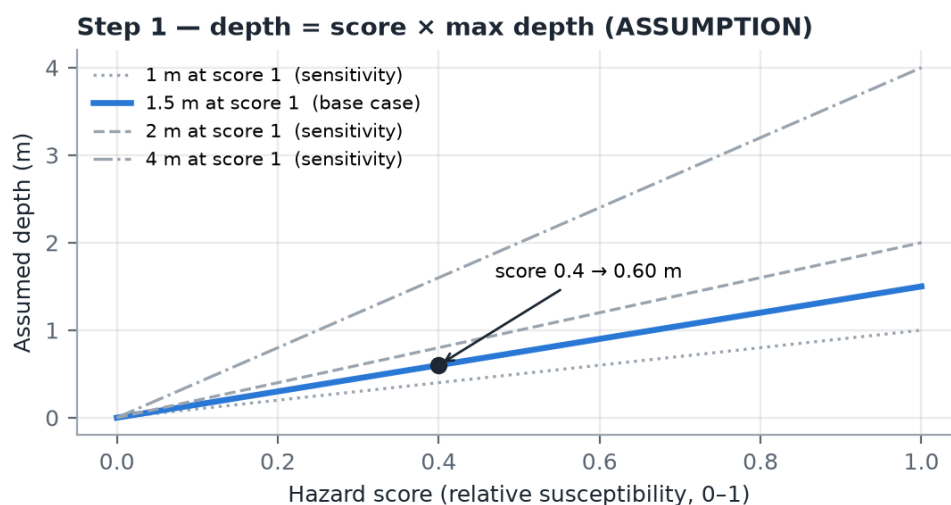


Figure 2. The base case and the sensitivity cases shown in the app. Changing the depth at score 1 from 1.5 m to 4 m multiplies every loss by about 2.4; to 1 m, by about 0.68.

**Basis:** our judgement of typical Nairobi street and household flooding. This is the weakest and most influential input: there is no citation for it yet. Observed depths (county or Red Cross assessments, or depths stated in flood reports) would firm it up.

## 3. Step 2 — depth to damage by construction class (ASSUMPTION)

**damage ratio = min( cap, JRC curve at depth ÷ class scale ).** A scale below 1 reaches a given damage at a shallower depth (more fragile); above 1, at a deeper one (more resilient).

| Class               | Depth scale | What it means                                                          | Cap | At score 1 |
|---------------------|-------------|------------------------------------------------------------------------|-----|------------|
| Informal iron-sheet | 0.5         | Same damage at half the JRC depth (light walls, contents on the floor) | 95% | 82.0%      |
| Semi-permanent      | 0.75        | Same damage at three-quarters of the depth                             | 90% | 64.0%      |
| Permanent masonry   | 1           | The JRC curve unchanged (closest to its source houses)                 | 85% | 53.0%      |
| Reinforced concrete | 1.3         | Needs 1.3× the depth (more resilient)                                  | 80% | 42.6%      |



Figure 3. Damage against hazard score for each class. Diamonds: the brief's reference dashboard at score 1 (85 / 69 / 52 / 31%). Ours: 82 / 64 / 53 / 43%. Caps never bind at 1.5 m.

**Basis:** the scales are construction reasoning, chosen as round numbers that keep the classes in order — no published source. The caps follow the brief's guidance that buildings rarely lose more than 80–95% of value because land and foundations survive. The reinforced-concrete scale moves this portfolio's loss most, because concrete buildings hold most of its value.

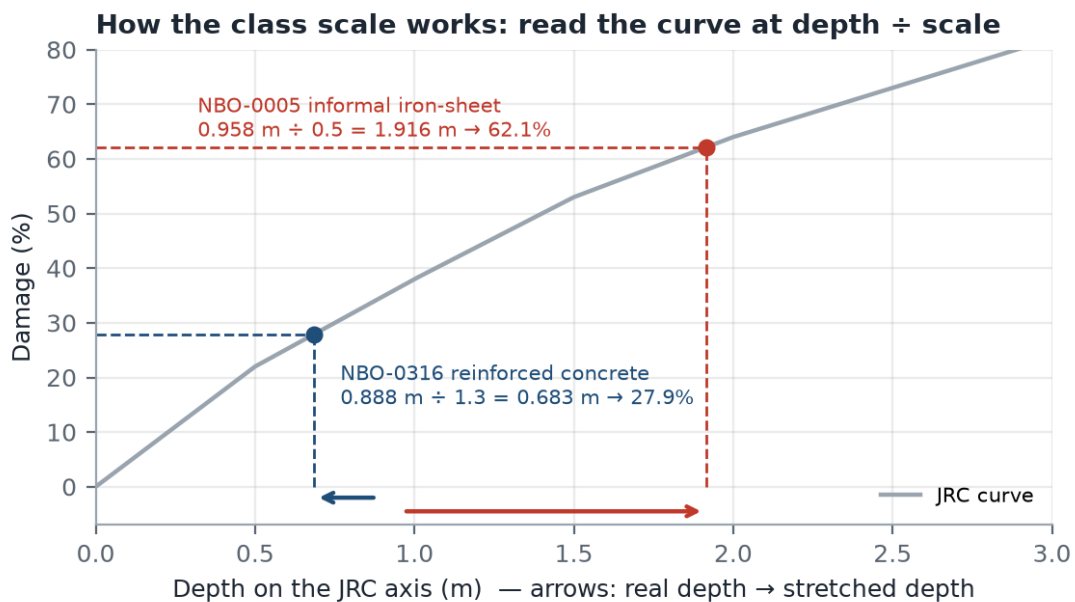


Figure 4. The same curve read at two stretched depths: the informal house is read further along the curve than its real depth, the concrete building less far.

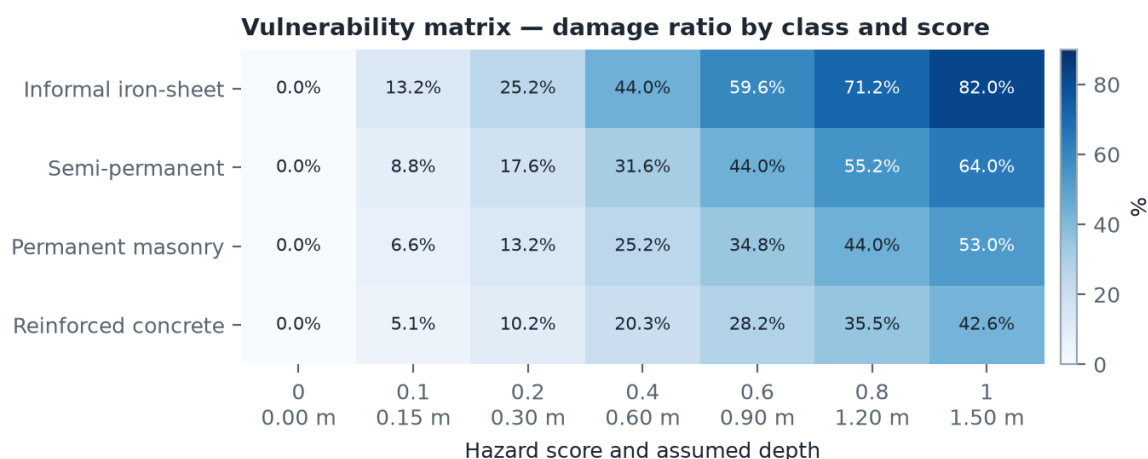


Figure 5. The vulnerability matrix the brief asks for. Each cell is checked by tests: 0 at no hazard, never decreasing, never above its cap, and fragile classes never below sturdier ones.

**One cell worked through** (score 0.4, depth 0.6 m): masonry reads the curve at 0.6 m, between 0.5 m (0.22) and 1.0 m (0.38):  $0.22 + (0.1 \div 0.5) \times 0.16 = \mathbf{25.2\%}$ . Informal reads it at  $0.6 \div 0.5 = 1.2$  m:  $0.38 + (0.2 \div 0.5) \times 0.15 = \mathbf{44.0\%}$ . Concrete reads it at  $0.6 \div 1.3 = 0.462$  m:  $(0.462 \div 0.5) \times 0.22 = \mathbf{20.3\%}$ .

## 4. Step 3 — damage to property loss

**loss = insured value × damage ratio × exposed share.** Insured value is the supplied `tiv_kes` (in the starter file it is about  $10 \times \text{floor area} \times \text{cost per m}^2$  in every row, cause unknown; kept as supplied and flagged). Exposed share applies only to multi-storey buildings with known storeys: (basements + 1 flooded storey) ÷ (storeys + basements), value spread evenly (ASSUMPTION). The starter portfolio has no storey data, so it is 1 here.

| Step                                  | NBO-0005 (informal iron-sheet)                                       | NBO-0316 (reinforced concrete)                                             |
|---------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------|
| Hazard score (1-in-100 tier)          | 0.6386                                                               | 0.5923                                                                     |
| Depth = score × 1.5 m                 | 0.958 m                                                              | 0.888 m                                                                    |
| Depth on the JRC axis = depth ÷ scale | $0.958 \div 0.5 = 1.916$ m                                           | $0.888 \div 1.3 = 0.683$ m                                                 |
| Damage ratio (cap not reached)        | <b>62.1%</b>                                                         | <b>27.9%</b>                                                               |
| Insured value                         | KES 1,570,000                                                        | KES 522,650,000                                                            |
| Loss                                  | $\text{KES } 1,570,000 \times 0.6215 = \mathbf{\text{KES } 975,726}$ | $\text{KES } 522,650,000 \times 0.2787 = \mathbf{\text{KES } 145,655,449}$ |

**Step 3 — loss = insured value × damage ratio (× exposed share = 1 here)**

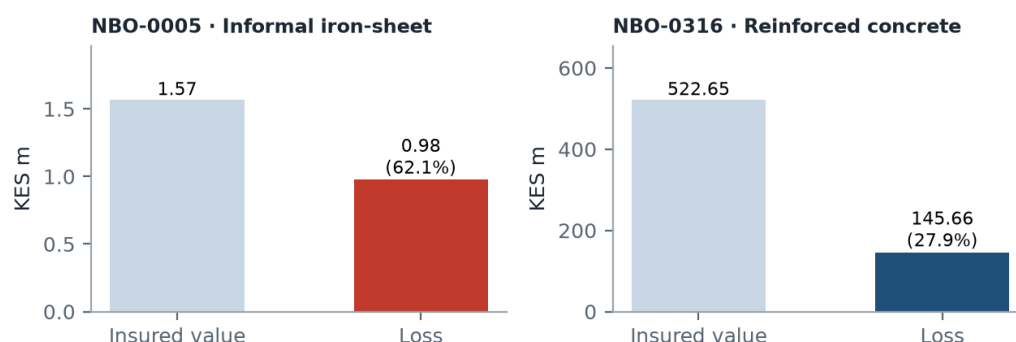


Figure 6. The concrete building takes the smaller share of damage but by far the larger loss, because it is worth hundreds of times more: value, not fragility, drives this portfolio.

This is the **ground-up loss**. The **gross loss** (on by default, ASSUMPTION) is  $\min(\text{limit}, \max(0, \text{ground-up} - \text{deductible}))$  per property, with a 1% deductible; the **net loss** is what remains after a quota share and a catastrophe excess of loss on each event; see the report §8.5b.

## 5. Steps 4-5 — portfolio loss by return period and average annual loss

Property losses are summed for each hazard tier, and each tier is given a return period (ASSUMPTION). The tiers are nested footprints: extreme flags only the most susceptible 5% of the map and is the frequent event; common flags 40% and is the rare one. The direction comes from the maps; the years are assumed.

| Tier       | Assumed return period | Annual chance | Portfolio loss | % of value |
|------------|-----------------------|---------------|----------------|------------|
| extreme    | 1-in-10               | 10.0%         | KES 0.26 bn    | 0.41%      |
| severe     | 1-in-25               | 4.0%          | KES 0.45 bn    | 0.70%      |
| moderate   | 1-in-50               | 2.0%          | KES 1.01 bn    | 1.59%      |
| occasional | 1-in-100              | 1.0%          | KES 1.70 bn    | 2.67%      |
| common     | 1-in-250              | 0.4%          | KES 2.65 bn    | 4.16%      |

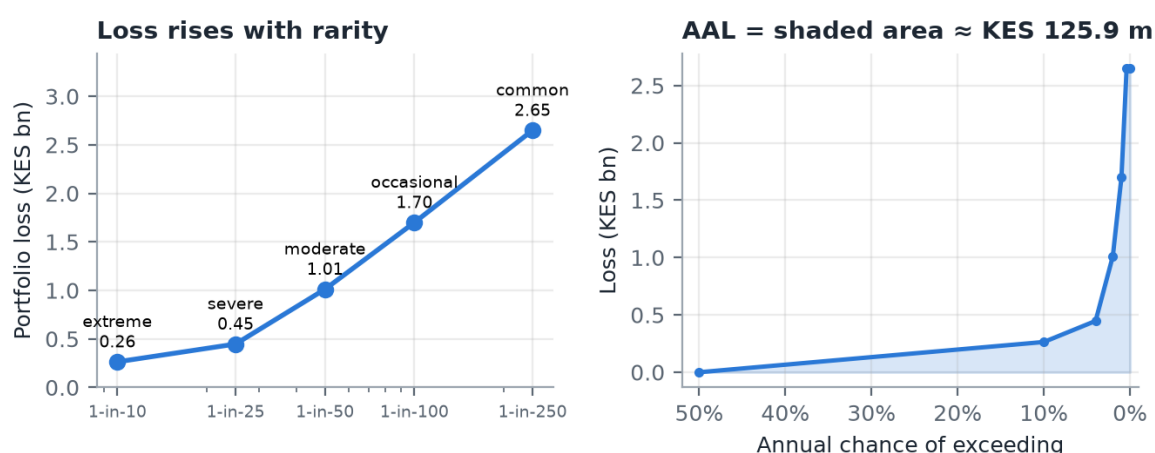
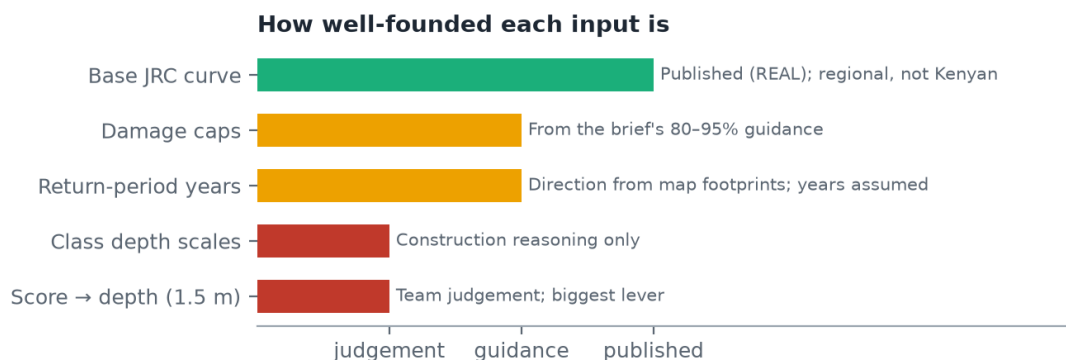


Figure 7. Left: the five scenario points. Right: loss against annual chance; the shaded area is the average annual loss.

| Range of annual chance     | Width × average loss                                                              | Contribution       |
|----------------------------|-----------------------------------------------------------------------------------|--------------------|
| 50.0% → 10.0%              | $0.400 \times (\text{KES } 0.00 \text{ bn} + \text{KES } 0.26 \text{ bn}) \div 2$ | KES 52.8 m         |
| 10.0% → 4.0%               | $0.060 \times (\text{KES } 0.26 \text{ bn} + \text{KES } 0.45 \text{ bn}) \div 2$ | KES 21.3 m         |
| 4.0% → 2.0%                | $0.020 \times (\text{KES } 0.45 \text{ bn} + \text{KES } 1.01 \text{ bn}) \div 2$ | KES 14.6 m         |
| 2.0% → 1.0%                | $0.010 \times (\text{KES } 1.01 \text{ bn} + \text{KES } 1.70 \text{ bn}) \div 2$ | KES 13.6 m         |
| 1.0% → 0.4%                | $0.006 \times (\text{KES } 1.70 \text{ bn} + \text{KES } 2.65 \text{ bn}) \div 2$ | KES 13.0 m         |
| 0.4% → 0 (held flat)       | $0.004 \times \text{KES } 2.65 \text{ bn}$                                        | KES 10.6 m         |
| <b>Average annual loss</b> |                                                                                   | <b>KES 125.9 m</b> |

Assumptions: no loss at 1-in-2 or more often; beyond the rarest tier the loss is held at its value. The app also simulates 10,000 years from these points for the full curve (log axis 1–10,000 years, with a sampling band).

## 6. Every assumption, and how well-founded it is



| Item                | Value                                | Basis                                                | Label      |
|---------------------|--------------------------------------|------------------------------------------------------|------------|
| Base damage curve   | JRC Africa residential, Table 3-1    | Published; regional, not calibrated to Kenya         | REAL       |
| Score → depth       | score × 1.5 m                        | Team judgement; biggest lever                        | ASSUMPTION |
| Class depth scales  | 0.5 / 0.75 / 1 / 1.3                 | Construction reasoning                               | ASSUMPTION |
| Damage caps         | 95% / 90% / 85% / 80%                | The brief's 80-95% guidance                          | ASSUMPTION |
| Loss basis          | damage × supplied insured value      | The brief's method; supplied value ≈ 10× area × cost | SYNTHETIC  |
| Flooded storeys     | basements + 1 storey                 | Value spread evenly by storey                        | ASSUMPTION |
| Return periods      | 10 / 25 / 50 / 100 / 250 years       | Reference mapping; direction from footprint size     | ASSUMPTION |
| Average annual loss | zero at 1-in-2; flat beyond 1-in-250 | Stated integration choices                           | ASSUMPTION |

**Where to strengthen it, in order:** (1) observed Nairobi flood depths for the score-to-depth conversion; (2) a published East African, material-based curve for the class scales — Englhardt et al. (2019, Natural Hazards and Earth System Sciences) built material-based curves for Ethiopia, to be checked for fit before use; (3) insurers' claims data, the only thing that would truly calibrate the curves.

Every value above lives in `configs/default.json`, is labelled in the app, and can be changed on the Assumptions page. This document is generated by `scripts/build_vulnerability_explainer.py` from the model itself, so it matches the code that produced it.